

energy harvester is to harness the energy and the most important ability of an energy harvester is power conditioning. Power conditioning is essential because the output from a transducer has low voltage

and current and might be at wrong frequency. Further, the intermittent output makes transducers unsafe for directly driving a device or charging a battery. The energy harvester IC conditions the input

power from the transducer and makes it usable.

Energy can be harnessed from different types of energy sources available in the surrounding environment. And the choice of energy harvesting technology depends on factors such as cost, power consumption, and availability near the device. Solar- and motion-driven are the most common types of energy harvesters while some others use thermal and radio frequency energy harvesting (RFHE).

Table 1 compares the power density of different ambient energy sources.

The different kind of energy harnessing methods—such as solar, thermal, and radio frequency energy harvesting—are briefly described below.

Solar energy harvesting

Solar energy harvesting is the most common type of ambient energy harvesting technique. The solar irradiance from the sun is readily available and the improvement of solar cells makes this a lucrative harvesting method in remote areas. ICs such as MAX20361 from Maxim Integrated and SPV1050 from STMicroelectronics utilise an external solar panel to harness solar energy into an electrical voltage. These ICs utilise MPPT to harness the maximum amount of energy. They also have either a buck or a buck-boost converter to step-up or step-down the energy to be utilised for charging the battery. This harvesting method is a great choice in remote areas that have lots of sunlight. The application circuit of SPV1050 is shown in Fig. 2.

The main considerations while selecting a solar energy harvester are the type of storage devices it can charge, its input voltage, minimum cold start voltage, etc. Table 2 compares two solar energy harvesters.

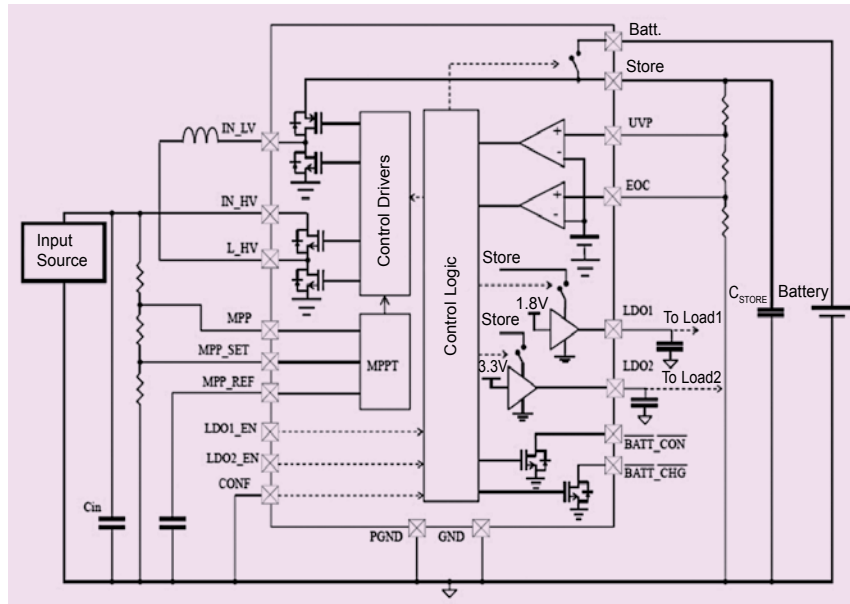


Fig. 2: Application circuit of SPV1050

Model	Manufacturer	Storage type it can charge	Energy can harness	Input voltage (minimum)	Output current	MPPT	Cold start
MAX 20361	Maxim Integrated	Rechargeable batteries, Capacitors	Solar	225mV	30mA	Yes	225mV
SPV 1050	ST Microelectronics	Rechargeable batteries, Capacitors	Thermal and solar	150mV	30-70mA	Yes	500mV (Boost conversion)

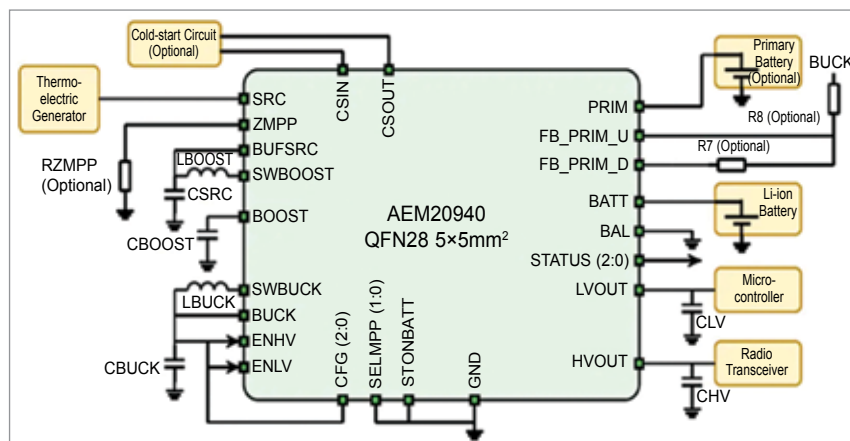


Fig. 3: Application circuit of e-Peas AEM20940